

MILLVILLE, NJ, USA

WMO: 724075

Lat: 39.366N

Lon: 75.078W

Elev: 18

StdP: 101.11

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13735

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-9.2	-19.7	0.7	-9.2	-17.1	0.8	-6.9	11.2	1.3	10.1	0.3	2.5	300	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	33.3	24.0	31.8	23.5	30.5	22.9	25.8	30.5	25.1	29.4	24.4	28.3	4.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.5	19.5	27.8	23.8	18.7	27.1	23.1	17.9	26.5	79.4	30.5	76.1	29.4	73.5	28.3	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.2	7.3	DB	-16.8	35.7	2.8	1.7	-18.9	36.9	-20.5	37.9	-22.1	38.8	-24.2	40.0	(4)
(5)			WB	-17.2	27.1	2.7	0.8	-19.1	27.7	-20.7	28.1	-22.3	28.6	-24.2	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	0.6	1.7	5.7	11.5	16.6	21.8	24.6	23.7	20.0	13.6	7.7	3.2
	DBStd	9.52	5.85	5.29	5.11	4.75	4.23	3.60	2.84	2.79	3.62	4.54	4.75	5.09
	HDD10.0	1057	296	237	153	36	2	0	0	0	0	16	100	218
	HDD18.3	2686	550	466	392	212	85	11	0	1	23	159	319	468
	CDD10.0	2016	5	5	19	81	208	355	453	424	300	127	31	8
	CDD18.3	604	0	0	1	7	33	116	194	166	74	12	1	0
	CDH23.3	5342	0	0	8	80	309	1031	1857	1441	540	75	2	0
	CDH26.7	1815	0	0	1	20	85	348	729	485	133	13	0	0
Wind	WSAvg	2.9	3.4	3.4	3.7	3.6	2.9	2.6	2.3	2.2	2.5	2.7	2.9	3.1
Precipitation	PrecAvg	1056	82	78	103	91	91	80	90	106	82	75	82	92
	PrecMax	1488	201	194	179	194	208	183	328	289	195	165	218	236
	PrecMin	610	5	16	32	14	20	11	14	22	15	2	13	11
	PrecStd	196	46	42	42	43	51	51	61	71	44	42	56	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	19.8	23.7	29.3	31.7	34.2	35.7	34.7	32.4	28.3	22.4	19.4
		MCWB	14.4	14.1	15.3	18.4	21.9	24.4	24.9	25.0	23.3	21.4	17.1	15.9
	2%	DB	15.1	15.6	20.0	25.7	29.1	32.1	33.6	32.4	29.9	25.4	19.6	16.1
		MCWB	13.1	11.4	13.4	16.8	20.8	23.2	24.2	24.0	22.5	20.2	15.2	13.8
	5%	DB	12.6	12.9	16.8	22.6	26.9	30.3	32.1	30.9	28.0	23.2	17.8	14.0
		MCWB	10.5	9.3	11.5	15.4	19.7	22.5	23.9	23.3	21.5	18.7	14.7	12.2
	10%	DB	9.7	10.2	14.1	19.9	24.6	28.6	30.6	29.3	26.4	21.4	16.0	11.7
		MCWB	7.4	7.0	9.9	13.9	18.5	21.8	23.2	22.6	20.7	17.5	13.2	9.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	15.5	17.1	20.2	23.3	25.5	26.8	26.9	25.2	23.2	18.7	17.1
		MCDB	16.8	18.1	22.5	25.9	29.5	31.3	32.1	31.6	29.1	25.3	20.7	18.7
	2%	WB	13.4	12.6	14.7	18.3	22.0	24.6	25.9	25.7	24.2	21.7	17.0	14.6
		MCDB	14.9	14.5	18.0	23.5	27.2	29.8	30.8	30.0	27.1	24.6	18.4	15.8
	5%	WB	10.7	10.0	12.8	16.9	20.9	23.9	25.2	24.9	23.4	20.1	15.4	12.5
		MCDB	12.2	12.2	15.5	21.1	25.3	28.5	29.8	28.6	26.2	22.2	17.2	13.7
	10%	WB	7.8	7.4	10.8	15.2	19.8	23.0	24.5	24.1	22.4	18.4	13.6	10.1
		MCDB	9.2	10.0	13.8	18.9	23.5	26.9	28.7	27.4	25.1	20.6	15.5	11.3
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.4	11.2	12.2	11.6	11.0	10.4	10.4	11.0	11.4	11.1	9.9
		MCDBR	12.2	13.7	15.0	15.3	13.8	12.4	11.7	11.5	11.9	12.7	12.4	11.5
		MCWBR	10.2	10.1	9.5	8.1	6.8	5.4	4.4	4.6	5.7	7.5	8.5	9.9
	5% WB	MCDBR	11.5	12.5	12.9	12.5	11.7	10.9	10.0	9.5	9.4	10.1	9.8	10.5
		MCWBR	10.5	10.6	9.7	7.6	6.5	5.2	4.4	4.3	5.1	6.9	8.1	9.9
Clear-Sky Solar Irradiance	taub	0.314	0.318	0.367	0.392	0.438	0.497	0.508	0.501	0.417	0.361	0.329	0.312	
	taud	2.472	2.458	2.362	2.322	2.223	2.104	2.087	2.093	2.302	2.463	2.510	2.523	
	Ebn at Noon	857	906	888	884	846	794	781	774	824	844	834	835	
	Edh at Noon	79	92	112	124	140	158	160	155	118	91	76	71	
All-Sky Solar Radiation	RadAvg	2.01	2.91	3.86	4.97	5.56	6.09	6.12	5.39	4.43	3.22	2.30	1.77	
	RadStd	0.17	0.32	0.31	0.39	0.67	0.47	0.28	0.35	0.44	0.32	0.20	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (72 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	+76

Nomenclature: See separate page